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Remotely operable welding machine controller device

Abstract

A remotely operable welding machine controller device for controlling a power output of an arc welding machine includes a housing and a remote control. A potentiometer and a power supply are attached to the housing. A motor is operationally engaged to a slider of the potentiometer and is positioned to selectively motivate the slider along turns of a resistance wire of the potentiometer to adjust a magnitude of current in a circuit in which the resistance wire is included. A power cord is attached to and extends from the housing. The power cord is electrically engaged to the potentiometer and operationally engages the potentiometer to a source of alternating current. The power supply is operationally engaged to the motor to selectively power the motor. The remote control is communicatively engaged to the motor is manipulatable by a user to selectively actuate the motor.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) Not Applicable

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

(2) Not Applicable

THE NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT

(3) Not Applicable

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISC
OR AS A TEXT FILE VIA THE OFFICE ELECTRONIC FILING SYSTEM

(4) Not Applicable

STATEMENT REGARDING PRIOR DISCLOSURES BY THE INVENTOR OR JOINT
INVENTOR

(5) Not Applicable

BACKGROUND OF THE INVENTION

(1) Field of the Invention

(6) The disclosure relates to welding machine controller devices and more particularly pertains to a new welding machine controller device for controlling a power output of an arc welding machine. The present invention discloses a welding machine controller device allowing remote control of

magnitude of a current supplied to an electrode.

(2) Description of Related Art Including Information Disclosed Under 37 CFR 1.97 and 1.98

(7) The prior art relates to welding machine controller devices, which may comprise secondary potentiometers, encoders that signal through welding wires, and wireless controllers. What is lacking in the prior art is a welding machine controller device comprising a motor for adjusting a potentiometer, wherein the motor is controlled via a wired or wireless remote.

BRIEF SUMMARY OF THE INVENTION

(8) An embodiment of the disclosure meets the needs presented above by generally comprising a housing, which defines an interior space, and a remote control. A potentiometer and a power supply are attached to the housing, with the potentiometer being positioned in the interior space. A motor is operationally engaged to a slider of the potentiometer and is positioned to selectively motivate the slider along turns of a resistance wire of the potentiometer to adjust a magnitude of current in a circuit in which the resistance wire is included. A power cord is attached to and extends from the housing. The power cord is electrically engaged to the potentiometer and is configured to operationally engage the potentiometer to a source of alternating current. The power supply is operationally engaged to the motor to selectively power the motor. The remote control is communicatively engaged to the motor and is configured to be manipulated by a user to selectively actuate the motor to adjust the magnitude of the current in the circuit.

(9) There has thus been outlined, rather broadly, the more important features of the disclosure in order that the detailed description thereof that follows may be better understood, and in order that the present contribution to the art may be better appreciated. There are additional features of the disclosure that will be described hereinafter and which will form the subject matter of the claims appended hereto.

(10) The objects of the disclosure, along with the various features of novelty which characterize the disclosure, are pointed out with particularity in the claims annexed to and forming a part of this disclosure.

Description

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWING(S)

(1) The disclosure will be better understood and objects other than those set forth above will become apparent when consideration is given to the following detailed description thereof. Such description makes reference to the annexed drawings wherein:

(2) FIG. 1 is a relay diagram of a remotely operable welding machine controller device according to an embodiment of the disclosure.

(3) FIG. 2 is an isometric perspective view of an embodiment of the disclosure.

(4) FIG. 3 is an isometric perspective view of an embodiment of the disclosure.

(5) FIG. 4 is an in-use view of an embodiment of the disclosure.

(6) FIG. 5 is a front view of an embodiment of the disclosure.

(7) FIG. 6 is an internal view of an embodiment of the disclosure.

DETAILED DESCRIPTION OF THE INVENTION

(8) With reference now to the drawings, and in particular to FIGS. 1 through 6 thereof, a new welding machine controller device embodying the principles and concepts of an embodiment of the disclosure and generally designated by the reference numeral 10 will be described.

(9) As best illustrated in FIGS. 1 through 6, the remotely operable welding machine controller device 10 generally comprises a housing 12, which defines an interior space 14, and a remote control 16, which may comprise an Up button 18 and a Down button 20, or other controlling means, such as, but not limited to, dials, touch panels, and the like. A potentiometer 22 and a power supply 24 are attached to the housing 12, with the former being positioned in the interior space 14.

A power cord **26** is attached to and extends from the housing **12**. The power cord **26** is electrically engaged to the potentiometer **22** and is configured to operationally engage the potentiometer **22** to a source of alternating current.

(10) A motor **28** is operationally engaged to a slider **30** of the potentiometer **22** and is positioned to selectively motivate the slider **30** along turns **32** of a resistance wire **34** of the potentiometer **22** to adjust a magnitude of current in a circuit in which the resistance wire **34** is included. The power supply **24** is operationally engaged to the motor **28** to selectively power the motor **28**. The power supply **24** comprises one or both of a power converter **36**, as shown in FIG. 5, or a battery (not shown). The power converter **36** is operationally engaged to the power cord **26** and may be external (as shown in FIG. 5) or internal (not shown) to the housing **12**.

(11) The remote control **16** is communicatively engaged to the motor **28** and is configured to be manipulated by a user to selectively actuate the motor **28** to adjust the magnitude of the current in the circuit. With an electrode holder included in the circuit, the user can perform welding distal from the housing **12** while controlling the current. The remote control **16** may be attached to the electrode holder by a variety of attachment means, such as, but not limited to, adhesives, tapes, zip ties, and the like. The present invention also anticipates the remote control **16** being integral to an electrode holder.

(12) A power and communications cable **38** is attached to and extends between the housing **12** and the remote control **16** so that the remote control **16** is communicatively engaged to the motor **28**. A first connector **40** is attached to the housing **12** and a second connector **42** is attached to the power and communications cable **38** distal from the remote control **16**. The second connector **42** is complementary to the first connector **40** and thus is selectively engageable to the first connector **40** for communicatively engaging the remote control **16** to the motor **28**. The present invention also anticipates a transmitter and a receiver (not shown), positioned in the remote control **16** and the housing **12**, respectively, thereby allowing for wireless control of the motor **28** by the remote control **16**.

(13) An amperage controller **44** is attached to the housing **12** and is operationally engaged to the potentiometer **22**. The amperage controller **44** is configured for manual adjustment of a setpoint of an amperage of the current in the circuit. The amperage controller **44** may comprise an Increase button **46** and a Decrease button **48**, as shown in FIG. 5, or other controlling means, such as, but not limited to, dials, touch panels, and the like.

(14) A pair of relays **50** is attached to the housing **12** and is positioned in the internal space. The relays **50** are operationally engaged to the power supply **24**, the motor **28**, and the remote control **16** so that the motor **28** is positioned to selectively increase or decrease the magnitude of the current in the circuit.

(15) A knob **52** is rotationally attached to the housing **12** and is operationally engaged to the slider **30**. The knob **52** is configured to be grasped in a hand of the user, positioning the user to adjust a voltage of the current in the circuit. A handle **54** is attached to a top **56** of the housing **12** and is configured to be grasped in a hand of the user to lift the housing **12**. A magnet **58** is attached to a bottom **60** of the housing **12** and is configured to magnetically attach the housing **12** to a ferromagnetic substrate. A set of outlets **62** is attached to the housing **12** and is operationally engaged to the power cord **26**. Each outlet **62** is configured for insertion of a plug to power a respective electronic device (not shown).

(16) In use, the remote control **16** and the electrode holder are grasped in a hand of a user and welding is performed. As may be required, the remote control **16** is used to adjust the magnitude of the current used in the welding process to obtain a high quality weld.

(17) With respect to the above description then, it is to be realized that the optimum dimensional relationships for the parts of an embodiment enabled by the disclosure, to include variations in size, materials, shape, form, function and manner of operation, assembly and use, are deemed readily apparent and obvious to one skilled in the art, and all equivalent relationships to those illustrated in

the drawings and described in the specification are intended to be encompassed by an embodiment of the disclosure.

(18) Therefore, the foregoing is considered as illustrative only of the principles of the disclosure. Further, since numerous modifications and changes will readily occur to those skilled in the art, it is not desired to limit the disclosure to the exact construction and operation shown and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the scope of the disclosure. In this patent document, the word “comprising” is used in its non-limiting sense to mean that items following the word are included, but items not specifically mentioned are not excluded. A reference to an element by the indefinite article “a” does not exclude the possibility that more than one of the element is present, unless the context clearly requires that there be only one of the elements.

Claims

1. A remotely operable welding machine controller device comprising: a housing defining an interior space; a potentiometer attached to the housing and positioned in the interior space; a power cord attached to and extending from the housing, the power cord being electrically engaged to the potentiometer, wherein the power cord is configured for operationally engaging the potentiometer to a source of alternating current; a motor positioned in the interior space and operationally engaged to a slider of the potentiometer, such that the motor is positioned for selectively motivating the slider along turns of a resistance wire of the potentiometer for adjusting a magnitude of current in a circuit in which the resistance wire is included; a power supply attached to the housing, the power supply being operationally engaged to the motor for selectively powering the motor; a remote control communicatively engaged to the motor, wherein the remote control is configured for being manipulated by a user for selectively actuating the motor for adjusting the magnitude of the current in the circuit; an amperage controller attached to the housing and operationally engaged to the potentiometer, wherein the amperage controller is configured for manual adjustment of a setpoint of an amperage of the current in the circuit; and a knob rotationally attached to the housing and operationally engaged to the slider, wherein the knob is configured for grasping in a hand of the user, positioning the user for adjusting a voltage of the current in the circuit.
2. The remotely operable welding machine controller device of claim 1, wherein the power supply comprises one or both of a power converter or a battery, the power converter being operationally engaged to the power cord.
3. The remotely operable welding machine controller device of claim 1, wherein the remote control comprises an Up button and a Down button.
4. The remotely operable welding machine controller device of claim 1, further including a power and communications cable attached to and extending between the housing and the remote control, such that the remote control is communicatively engaged to the motor.
5. The remotely operable welding machine controller device of claim 4, further including: a first connector attached to the housing; and a second connector attached to the power and communications cable distal from the remote control, the second connector being complementary to the first connector, such that the second connector is selectively engageable to the first connector for communicatively engaging the remote control to the motor.
6. The remotely operable welding machine controller device of claim 1, wherein the amperage controller comprises an Increase button and a Decrease button.
7. The remotely operable welding machine controller device of claim 1, further including a pair of relays attached to the housing and positioned in the internal space, the relays being operationally engaged to the power supply, the motor, and the remote control, such that the motor is positioned for selectively increasing or decreasing the magnitude of the current in the circuit.
8. The remotely operable welding machine controller device of claim 1, further including a handle

attached to a top of the housing, wherein the handle is configured for grasping in a hand of the user for lifting the housing.

9. The remotely operable welding machine controller device of claim 1, further including a magnet attached to a bottom of the housing, wherein the magnet is configured for magnetically attaching the housing to a ferromagnetic substrate.

10. The remotely operable welding machine controller device of claim 1, further including a set of outlets attached to the housing and operationally engaged to the power cord, wherein each outlet is configured for insertion of a plug for powering a respective electronic device.

11. A remotely operable welding machine controller device comprising: a housing defining an interior space; a potentiometer attached to the housing and positioned in the interior space; a power cord attached to and extending from the housing, the power cord being electrically engaged to the potentiometer, wherein the power cord is configured for operationally engaging the potentiometer to a source of alternating current; a motor positioned in the interior space and operationally engaged to a slider of the potentiometer, such that the motor is positioned for selectively motivating the slider along turns of a resistance wire of the potentiometer for adjusting a magnitude of current in a circuit in which the resistance wire is included; a power supply attached to the housing, the power supply being operationally engaged to the motor for selectively powering the motor, the power supply comprising one or both of a power converter or a battery, the power converter being operationally engaged to the power cord; a remote control communicatively engaged to the motor, wherein the remote control is configured for being manipulated by a user for selectively actuating the motor for adjusting the magnitude of the current in the circuit, the remote control comprising an Up button and a Down button; a power and communications cable attached to and extending between the housing and the remote control, such that the remote control is communicatively engaged to the motor; a first connector attached to the housing; a second connector attached to the power and communications cable distal from the remote control, the second connector being complementary to the first connector, such that the second connector is selectively engageable to the first connector for communicatively engaging the remote control to the motor; an amperage controller attached to the housing and operationally engaged to the potentiometer, wherein the amperage controller is configured for manual adjustment of a setpoint of an amperage of the current in the circuit, the amperage controller comprising an Increase button and a Decrease button; a pair of relays attached to the housing and positioned in the interior space, the relays being operationally engaged to the power supply, the motor, and the remote control, such that the motor is positioned for selectively increasing or decreasing the magnitude of the current in the circuit; a knob rotationally attached to the housing and operationally engaged to the slider, wherein the knob is configured for grasping in a hand of the user, positioning the user for adjusting a voltage of the current in the circuit; a handle attached to a top of the housing, wherein the handle is configured for grasping in a hand of the user for lifting the housing; a magnet attached to a bottom of the housing, wherein the magnet is configured for magnetically attaching the housing to a ferromagnetic substrate; and a set of outlets attached to the housing and operationally engaged to the power cord, wherein each outlet is configured for insertion of a plug for powering a respective electronic device.
